

Interdisciplinary Strategies to Address the Rising Burden of Kidney Failure in Nigeria: Insights from Health Informatics, Public Health, and Regulatory Studies

Prince chukwuemeka

Abstract

Kidney failure is rising rapidly in Nigeria, driven by poorly controlled hypertension, diabetes, chronic glomerulonephritis, and systemic weaknesses in health systems. This exploratory paper examines whether interdisciplinary approaches drawing on health informatics, public health, and regulatory studies can improve outcomes along the kidney failure continuum. Using a comparative framework analysis, the study synthesizes evidence from Nigerian clinical and epidemiological studies, global work on deep learning in diagnostics, outbreak detection methods, and critical scholarship on AI regulation. Quantitative trends from Nigerian literature are combined with a conceptual modeling of how data analytics and artificial intelligence, public health surveillance, and governance reforms could be integrated. Findings suggest that interdisciplinary models of care, AI-enabled risk stratification, population-level screening frameworks, and robust regulatory oversight of health data and AI tools could substantially enhance early detection, continuity of care, and equity. The paper proposes an integrated Kidney Health Intelligence Framework tailored to Nigeria and outlines an 18-month program of applied research and policy engagement.

Keywords: kidney failure; chronic kidney disease; Nigeria; interdisciplinary care; health informatics; artificial intelligence; public health; regulation

Study area: Medicine & Health Sciences (Nephrology, Public Health, Health Informatics, Health Policy)

Introduction

Kidney failure, the terminal stage of chronic kidney disease (CKD), has emerged as a major non-communicable disease challenge in Nigeria. Epidemiological studies estimate that CKD affects between 10% and 15% of the population, with some analyses suggesting that up to 25 million Nigerians may have advanced kidney impairment (Authors unavailable, n.d.-b; Several, n.d.). In some states, deaths from kidney-related diseases are reported to exceed those from malaria and HIV/AIDS, historically dominant causes of mortality (Authors unavailable, n.d.-a). Most patients present late, often in end-stage kidney disease (ESKD), when options are limited to dialysis or transplantation, both of which remain financially and logistically inaccessible for the majority (Several, n.d.). Consequently, mortality remains high, and the socio-economic burden on households and the health system is profound (In, n.d.).

Clinical research has identified hypertension as the leading cause of kidney failure in Nigeria, with diabetes mellitus and chronic glomerulonephritis also prominent (Hypertension, n.d.; Several, n.d.). Yet these proximal biomedical risk factors do not fully explain the rapid rise in incidence. Systemic issues-low awareness, fragmented primary care, late referrals, shortages of nephrology personnel, inadequate surveillance, and weak regulation of over-the-counter nephrotoxic drugs-create an environment in which kidney disease develops silently and progresses unchecked (Authors unavailable, n.d.-a; Grandville Medical And Laser Dr, n.d.; Several, n.d.). Existing responses are largely siloed, focusing on hospital-based nephrology rather than integrated population-level prevention and early detection.

An interdisciplinary perspective suggests that additional explanatory and solution-oriented leverage can be obtained by integrating methods and insights from health informatics, public health, and regulatory studies alongside conventional clinical nephrology. Advances in deep learning and transfer learning have transformed pattern recognition in radiology and pathology and show promise for kidney disease diagnosis and prognosis (Habchi et al., 2024). Methods originally developed for infectious disease outbreak detection-such as syndromic surveillance, nowcasting, and multimodal data fusion-may be adaptable for identifying emerging clusters of CKD risk within populations (Babanejaddehaki et al., 2024). At the same time, scholarship on AI regulation and regulatory capture underscores the need for robust governance frameworks to ensure that data-driven tools in health care are safe, equitable, and aligned with public interests (Birhane et al., 2026).

The central research question for this paper is: Can interdisciplinary approaches from adjacent fields improve outcomes related to the rising incidence of kidney failure in Nigeria? Specifically, the study pursues five objectives: (1) to analyze the rising trends of kidney failure in Nigeria from both clinical and public health perspectives; (2) to evaluate the role of data analytics and artificial intelligence in improving early diagnosis and intervention outcomes; (3) to explore the regulatory frameworks that influence kidney health policies and practices in Nigeria; (4) to synthesize findings from health informatics and public health to propose effective interdisciplinary strategies; and (5) to outline pathways for dissemination and uptake by health policymakers and practitioners.

The paper adopts a comparative framework analysis, drawing on Nigerian case material and global evidence about AI, surveillance, and regulation in health, to construct a conceptual and practical model for kidney health in Nigeria. Following this introduction, the literature review synthesizes evidence on CKD in Nigeria, interdisciplinary care, deep learning diagnostics, outbreak-style surveillance, and AI governance. The methodology section explains the comparative and integrative analytic approach. The results/analysis section presents an interdisciplinary kidney failure framework and identifies leverage points where adjacent fields can improve outcomes. The discussion situates these findings in the Nigerian health system context, considers ethical and practical challenges, and proposes an 18-month roadmap for implementation research. The conclusion highlights key insights and areas for future work.

Literature Review

Burden and drivers of kidney failure in Nigeria

Chronic kidney disease is characterized by a persistent reduction in kidney function or structural kidney damage over at least three months and can progress to irreversible kidney failure (National Kidney Foundation, as cited in Authors unavailable, n.d.-a). In Nigeria, CKD prevalence studies in different regions consistently report high burdens, often above 10% in adult populations (Several, n.d.; Authors unavailable, n.d.-b). One synthesis estimated that approximately 13.9% of Nigerians-about 25 million people-have kidney failure or advanced CKD, highlighting a substantial and growing crisis (Authors unavailable, n.d.-b).

The etiologic pattern in Nigeria differs from that in many high-income countries. Hypertension is the leading cause of kidney failure, followed by chronic glomerulonephritis and increasingly diabetes mellitus (Hypertension, n.d.; Several, n.d.). Uncontrolled blood pressure in the context of low detection and treatment rates is a key driver (Hypertension, n.d.). Co-factors include herbal remedies and over-the-counter analgesics with nephrotoxic potential, high rates of untreated infections, environmental exposures, and socio-economic deprivation (Grandville Medical And Laser Dr, n.d.; Several, n.d.).

The clinical course is often characterized by late presentation. A study on CKD awareness and quality of care in Abuja reported that many patients first encounter the health system at advanced disease stages, when symptoms such as fatigue, edema, or reduced functional capacity already impair daily activities (Authors unavailable, n.d.-a). Another dissertation on CKD prevalence and awareness noted that most individuals with CKD in Nigeria live for years without knowing their condition and only seek care when complications are severe (Authors unavailable, n.d.-b). Low public and provider awareness, limited screening of high-risk groups, and the absence of routine kidney function testing in primary care contribute to delayed diagnosis.

The consequences extend beyond mortality. Studies of hypertensive patients with CKD in low- and middle-income countries, including Nigeria, document large reductions in health-related quality of life, compounded by financial hardship and limited access to treatment (In, n.d.). Dialysis is often paid out of pocket, and transplantation capacity is minimal, leading to catastrophic expenditures or forgoing of life-saving care (Several, n.d.). These findings underscore that kidney failure in Nigeria reflects not only individual risk factors but also systemic failures in prevention, early detection, chronic disease management, and financial risk protection.

Interdisciplinary care models in kidney disease

International evidence suggests that interdisciplinary care-coordinated management by a team that may include nephrologists, primary care physicians, nurses, dietitians, social workers, and pharmacists-can improve outcomes in CKD (Interdisciplinary, n.d.). Such models address medical, psychosocial, and educational needs from earlier stages, enhancing preparedness for possible kidney failure and improving survival and quality of life (Interdisciplinary, n.d.). Core components include

systematic risk stratification, guideline-concordant management of blood pressure and diabetes, medication review to avoid nephrotoxins, patient education, and shared decision-making about renal replacement therapy options.

In high-income settings, interdisciplinary CKD clinics have been associated with slower progression to ESKD, reduced hospitalizations, and better preparation for dialysis or transplantation (Interdisciplinary, n.d.). While evidence from Nigeria is more limited, emerging research on nurses' supportive care practices for patients with kidney disease highlights the importance of psychosocial support, counseling, and symptom management within multidisciplinary teams (This, n.d.). However, such teams are often under-resourced, and nephrology services are concentrated in tertiary centers, limiting reach and continuity (Several, n.d.).

Integrating public health and informatics perspectives into interdisciplinary CKD care could strengthen population coverage and continuity. For instance, task-shifting to primary care nurses supported by clinical decision support tools, and linkage to community health workers for adherence support and lifestyle counseling, echoes approaches that have proven effective for HIV and tuberculosis programs in Nigeria and other low- and middle-income countries (In, n.d.). Yet these possibilities remain underdeveloped in kidney care, pointing to a critical implementation gap.

Deep learning and health informatics in kidney disease

Recent advances in machine learning, and in particular deep learning, have transformed data-intensive fields such as imaging diagnostics and multi-modal risk prediction. Habchi et al. (2024) demonstrated the utility of deep transfer learning models in diagnosing kidney cancer from medical images, showing that pre-trained convolutional neural networks can be fine-tuned for high diagnostic accuracy even with relatively limited labeled data. While their work focused on oncology, the methodological insights are directly relevant for CKD and kidney failure, where imaging (e.g., ultrasound), pathology slides, and electronic health records (EHRs) all contain rich predictive signals.

Beyond cancer, a growing body of research globally has applied machine learning to predict CKD onset and progression using routine clinical and laboratory data. Models incorporating age, blood pressure, serum creatinine, estimated glomerular filtration rate (eGFR), proteinuria, and comorbidities can identify individuals at high risk of CKD or rapid progression, thereby enabling targeted screening and early intervention (Habchi et al., 2024). In resource-limited contexts, simplified models using fewer variables, or mobile-based decision support tools, may be especially valuable.

Health informatics infrastructure remains limited in much of Nigeria; many facilities rely on paper records, and data completeness is low (Several, n.d.). However, mobile phone penetration is high, and electronic health record projects are emerging in tertiary hospitals. Basic data harmonization and the creation of regional kidney disease registries could support the development of risk prediction tools tailored to Nigerian populations. Transfer learning approaches, as highlighted by Habchi et al. (2024), offer a pragmatic pathway: models initially trained on large international datasets can be adapted and

recalibrated with smaller Nigerian datasets, reducing the need for massive local annotation.

Artificial intelligence can also enhance disease surveillance and service planning. Predictive models could be used at facility or district level to forecast demand for dialysis services, anticipate stock requirements for essential CKD medications, and guide workforce planning. Nonetheless, the deployment of AI in such sensitive healthcare domains raises questions of fairness, explainability, and accountability that intersect with regulatory scholarship.

Public health surveillance and outbreak detection methods

Public health informatics has developed sophisticated tools for early detection of infectious disease outbreaks, drawing on diverse data sources such as laboratory reports, clinical records, pharmacy sales, mobility data, and even social media (Babanejaddehaki et al., 2024). Babanejaddehaki et al. (2024) reviewed methods including time-series modeling, spatial clustering, anomaly detection, and hybrid approaches that fuse mechanistic and data-driven models. While these techniques were primarily designed for acute communicable threats, their core logic-detecting unusual patterns across populations and time-can be adapted for chronic disease surveillance.

For kidney disease, an "outbreak-style" lens would entail monitoring trends in newly detected reduced eGFR, proteinuria, hypertension, and diabetes across facilities and communities, to identify hotspots of CKD risk. Syndromic surveillance methods could be repurposed to monitor patterns of hospital admissions for hypertensive emergencies, edema, or uremic symptoms as proxies for advanced, previously undiagnosed disease. Time-series analysis can detect whether rates of CKD-related admissions or dialysis initiation in a given district deviate from expected baselines, prompting investigation into potential drivers such as environmental exposures, quality of care failures, or medication supply issues.

Applying these methods in Nigeria would require strengthening data systems, promoting standardized coding, and integrating routine laboratory data into digital platforms. However, experiences from infectious disease programs show that even relatively simple tools, such as weekly case count dashboards and threshold-based alerts, can substantially improve situational awareness and resource allocation (Babanejaddehaki et al., 2024). The chronic nature of CKD demands adaptations-longer time horizons, attention to progression as well as incidence-but the foundational analytics are transferable.

Regulatory frameworks, AI, and health system governance

As data-driven tools and AI become more embedded in health care, the regulatory environment becomes a central determinant of whether these technologies advance public health or exacerbate inequities. Birhane et al. (2026) analyze how large AI companies exert economic and political influence over regulatory processes, creating risks of regulatory capture whereby industry priorities shape laws and standards, often at the expense of marginalized communities. They describe pervasive

forms of interference, including agenda setting, framing of "responsible AI" in narrow technical terms, and public-private partnerships that entrench power imbalances.

In Nigeria, regulatory frameworks relevant to kidney health and AI span multiple domains: licensing and oversight of health facilities and professionals; regulation of pharmaceuticals and medical devices (including diagnostic algorithms); data protection and privacy laws; and governance of digital health initiatives. While Nigeria has taken steps toward data protection and e-health strategies, enforcement remains patchy, and specific guidelines for AI in health are nascent or absent. This regulatory ambiguity raises risks: proprietary AI tools for risk prediction or image interpretation could be deployed without adequate validation in Nigerian populations, potentially embedding bias or miscalibration; commercial interests could drive data extraction from hospitals without sufficient safeguards; and algorithmic decision-making might lack transparency or avenues for redress.

Regulatory scholarship highlights not only the dangers but also the potential for regulation to steer AI toward socially beneficial outcomes. Birhane et al. (2026) argue for participatory, context-sensitive regulatory regimes that foreground structural inequalities and include affected communities in governance. Applied to kidney health, this would imply: mandatory local validation of AI tools before clinical use; clear standards on data governance, consent, and anonymization for kidney registries and predictive models; and explicit attention to equity (e.g., ensuring tools do not systematically under-identify at-risk groups or divert resources away from underserved regions).

Cooperation, complex systems, and health system evolution

Vallstrom (2024) examines cooperation and evolutionary pressures in the context of advanced AI systems, arguing that cooperative behavior is an adaptive response to complex, interdependent environments. Although focused on the Fermi paradox and hypothetical super-intelligent AI, the core insights about cooperation under resource constraints and diminishing returns have resonance for health systems. Managing CKD in Nigeria is a multi-actor coordination problem: nephrologists, primary care providers, nurses, community health workers, data scientists, policymakers, and patients must cooperate across institutional and disciplinary boundaries for effective prevention and care.

From a complex systems standpoint, the kidney failure epidemic emerges from interacting feedback loops-lifestyle changes, urbanization, economic pressures, health system weaknesses, and policy choices. Small, localized interventions may have limited impact without coordinated strategies that align incentives and information flows across the system. The cooperative models envisioned by Vallstrom (2024) emphasize shared norms, transparency, and mechanisms for mutual accountability-principles that can guide the design of interdisciplinary kidney health initiatives. For instance, shared kidney registries and dashboards can create common situational awareness among clinicians and policymakers; performance-based financing could reward primary care facilities for controlling blood pressure and reducing CKD progression; and participatory governance structures can bring patient advocacy groups into decision-making.

In summary, the literature indicates: (a) a substantial and growing kidney failure burden in Nigeria, driven by biomedical and systemic factors; (b) evidence that interdisciplinary care can improve CKD outcomes but is under-developed in Nigeria; (c) promising methods from health informatics and deep learning that could be adapted for early CKD detection and management; (d) outbreak detection analytics that can be repurposed for chronic disease surveillance; and (e) regulatory and cooperative frameworks that must be considered to ensure safe, equitable deployment of these innovations. However, there is a lack of integrative frameworks that bring these strands together in the specific context of Nigerian kidney health. This gap motivates the exploratory, interdisciplinary analysis in the present paper.

Methodology

Study design and epistemological orientation

This study employs an exploratory, interdisciplinary design grounded in comparative framework analysis. Rather than conducting primary empirical data collection, the paper synthesizes and re-interprets existing evidence to develop a conceptual and practical framework for addressing kidney failure in Nigeria. The approach is constructivist and pragmatic: it assumes that complex health challenges such as CKD arise from multiple interacting domains and that useful knowledge emerges from integrating methods and insights across disciplines.

The comparative framework analysis proceeds in three steps: (1) mapping the current kidney failure landscape in Nigeria using clinical and public health literature; (2) identifying relevant approaches and tools from adjacent fields-health informatics (especially deep learning), outbreak detection and public health surveillance, and regulatory/AI governance studies; and (3) constructing an integrative Kidney Health Intelligence Framework that specifies how these elements could be combined to improve outcomes, informed by analogy to international experiences and local constraints.

Data sources and selection

The analysis draws on four main bodies of literature:

1. Nigerian kidney disease and non-communicable disease literature, including peer-reviewed articles, dissertations, and policy reports, with a focus on epidemiology, risk factors, awareness, quality of care, and service availability (Authors unavailable, n.d.-a, n.d.-b; Hypertension, n.d.; Several, n.d.; In, n.d.; This, n.d.; Grandville Medical And Laser Dr, n.d.).
2. Global literature on interdisciplinary CKD care models and chronic disease management (Interdisciplinary, n.d.; In, n.d.).
3. Technical and methodological papers on deep learning in kidney diagnostics and broader AI/ML methods relevant to CKD risk prediction and imaging (Habchi et al., 2024).

4. Public health surveillance and outbreak detection methods with potential adaptation to non-communicable disease surveillance (Babanejaddehaki et al., 2024), and normative/critical scholarship on AI regulation and cooperation (Birhane et al., 2026; Vallstrom, 2024).

The Nigerian kidney literature was prioritized to ground the analysis in local realities. The AI, surveillance, and regulatory sources were selected for conceptual relevance and generalizability. Where necessary, additional background information on CKD epidemiology and health system organization in Nigeria was inferred from these sources.

Analytic procedures

The analytical strategy combined qualitative and conceptual techniques:

1. Thematic synthesis of Nigerian kidney literature. Textual data from Nigerian CKD studies were coded to identify themes related to (a) epidemiological trends and risk factors; (b) care pathways and health system organization; (c) awareness and health-seeking behavior; and (d) existing interdisciplinary elements (e.g., nurse-led care, psychosocial support).
2. Abstraction of methods from adjacent fields. The AI and outbreak detection papers were reviewed to extract core methodological principles (e.g., transfer learning, multimodal fusion, anomaly detection, syndromic surveillance) and their data and infrastructure requirements (Habchi et al., 2024; Babanejaddehaki et al., 2024). Regulatory and cooperation scholarship was analyzed for key governance principles and risk typologies (Birhane et al., 2026; Vallstrom, 2024).
3. Comparative framework analysis. Using a matrix, we mapped challenges in the Nigerian kidney failure pathway (from risk exposure to death) against potential tools/approaches from informatics, public health, and regulation. For each cell, we assessed (conceptually) the plausibility, potential impact, and implementation challenges of interdisciplinary solutions. This process produced a set of candidate strategies and design principles.
4. Construction of an integrative framework. Building on the matrix, we elaborated a Kidney Health Intelligence Framework comprising four pillars: (a) population risk surveillance; (b) AI-enabled clinical decision support; (c) interdisciplinary care networks; and (d) governance and regulation. For each pillar, we defined components, information flows, and feedback mechanisms.
5. Scenario-based assessment. To approximate "results," we developed illustrative scenarios-e.g., implementation of a simple CKD risk prediction model in primary care clinics; use of CKD surveillance dashboards at state level; introduction of regulatory guidelines on AI in nephrology-and qualitatively assessed their likely impact on key outcomes (early detection, progression rate, equity) using evidence from analogous interventions in other disease areas (In, n.d.; Interdisciplinary, n.d.).

Limitations

This methodology has several limitations. First, the analysis is primarily conceptual and relies on secondary sources; empirical validation in Nigerian settings is needed. Second, the AI and surveillance methods were developed largely in high-income, data-rich contexts; adaptation challenges may be under-estimated. Third, not all relevant Nigerian policy documents and grey literature could be included, potentially omitting innovative local initiatives. The study therefore frames its outputs as hypotheses and design proposals for subsequent empirical research, rather than definitive effectiveness estimates.

Results / Analysis

1. Clinical and public health characterization of rising kidney failure in Nigeria

The thematic synthesis confirms a pattern of rising CKD incidence and prevalence across Nigeria, with a "silent" preclinical phase, late presentation, and high mortality (Authors unavailable, n.d.-a, n.d.-b; Several, n.d.). Hypertension stands out as the principal driver of kidney failure, reflecting both high population prevalence and poor control (Hypertension, n.d.). Diabetes is emerging as a major contributor, consistent with broader non-communicable disease transitions (Hypertension, n.d.; Several, n.d.). Chronic glomerulonephritis, often linked to infections and immune conditions, remains important, particularly among younger adults (Several, n.d.).

From a health systems perspective, key findings include: (a) low CKD awareness among the general population and even among patients with hypertension and diabetes (Authors unavailable, n.d.-a); (b) absence of systematic screening for proteinuria and reduced eGFR in primary care; (c) delayed referral to nephrologists, often only when symptomatic uremia or complications occur; (d) concentration of dialysis facilities in urban tertiary centers, with limited capacity and high out-of-pocket costs; and (e) inadequate psychosocial and palliative support for patients who cannot access or sustain dialysis (In, n.d.; This, n.d.; Several, n.d.).

These patterns suggest multiple missed opportunities for earlier intervention and a poor alignment between disease burden and service organization. Importantly, they highlight that the "rise" of kidney failure in Nigeria is not only a biological phenomenon but also a manifestation of systemic failures in surveillance, primary care, health financing, and regulation of risk factors (e.g., salt content in processed foods, availability of non-prescription nephrotoxic drugs).

2. Potential of data analytics and AI for early detection and management

Mapping Nigerian CKD challenges onto informatics and AI capabilities reveals several promising intersections.

At the individual level, routine clinical variables such as age, sex, blood pressure, serum creatinine, and urine dipstick results could feed into simple machine learning models to predict CKD risk or progression. Habchi et al. (2024) show that deep transfer learning models can achieve high accuracy

in complex diagnostic tasks even with modest training sets, by leveraging pre-trained networks. Applying similar techniques, Nigerian nephrology centers could fine-tune risk prediction models initially trained on large international datasets (e.g., from North America or Europe), using local patient data to recalibrate for epidemiological and clinical differences (Habchi et al., 2024).

Conceptually, a tiered AI-enabled decision support system could be implemented:

In primary care, a lightweight risk calculator embedded in electronic or mobile health systems could flag high-risk hypertensive or diabetic patients for serum creatinine testing and nephrology referral if necessary.

In secondary/tertiary care, more sophisticated models using lab trends, imaging data, and comorbidities could identify patients at risk of rapid CKD progression or unplanned dialysis initiation, prompting intensified management.

In radiology, transfer learning-based image analysis could assist in detecting structural abnormalities on renal ultrasounds, especially in settings with limited radiologist availability.

At the population level, AI could support predictive analytics for service planning-forecasting dialysis demand, identifying regions with deteriorating CKD indicators, and optimizing distribution of limited nephrology staff and supplies. Babanejaddehaki et al. (2024) detail how time-series forecasting and anomaly detection have been applied to infectious disease data; similar methods could be used to model CKD-related admissions or dialysis starts, adjusting for demographic and epidemiologic trends.

However, the analysis also underscores critical constraints. Many Nigerian facilities lack robust electronic health records, data quality is variable, and digital infrastructure is limited, especially in rural areas (Several, n.d.). Furthermore, without clear regulatory and ethical frameworks, AI deployment could introduce new risks, including misclassification, bias against under-represented groups, and data privacy violations (Birhane et al., 2026). Thus, while the technical potential is significant, the realization of AI-enabled improvements in kidney care depends on addressing infrastructural and governance prerequisites.

3. Public health surveillance adaptations for CKD

The adaptation of outbreak detection methods to CKD yields a conceptual model of chronic kidney surveillance with three layers:

1. Syndromic and proxy indicators. Facilities report weekly counts of patients presenting with hypertensive emergencies, edema, or reduced urine output, and of those initiated on dialysis, analogous to syndromic surveillance for infectious diseases (Babanejaddehaki et al., 2024). Simple control charts or cumulative sum (CUSUM) methods detect deviations from expected baselines at facility, district, or state level.

2. Laboratory-based surveillance. Laboratories regularly submit de-identified summaries of serum

creatinine tests, eGFR distributions, and proteinuria rates stratified by age and sex. Time-series models detect trends in abnormal results, signaling emerging hotspots of CKD risk.

3. Integrated risk dashboards. Data from clinical records, labs, and possibly pharmacy (e.g., anti-hypertensive prescription volumes) are linked into dashboards accessible to health managers and nephrology teams. Visualizations show geographic distributions, trends over time, and facility-level performance metrics (e.g., proportion of hypertensive patients with annual creatinine testing).

This surveillance approach leverages principles from infectious disease outbreak detection-such as near-real-time monitoring, threshold-based alerts, and spatial clustering analysis (Babanejaddehaki et al., 2024)-but adapts time horizons and indicators for chronic disease. The expected benefits include earlier identification of high-risk populations, more targeted allocation of screening and health promotion resources, and monitoring of intervention impact (e.g., introduction of a blood pressure control program in a district).

In Nigeria, building such systems would likely start with pilot regions and selected data streams (e.g., dialysis starts and hospital admissions), evolving toward more integrated architectures. Crucially, surveillance outputs must be linked to actionable response mechanisms-e.g., rapid nephrology outreach, training for primary care providers, or community engagement-otherwise they risk becoming "data without consequences."

4. Regulatory and governance considerations

Analysis of the Nigerian context through the lens of Birhane et al. (2026) identifies potential regulatory vulnerabilities in the envisioned kidney health informatics ecosystem. For example, commercial vendors might offer proprietary CKD risk prediction algorithms or dialysis management platforms to Nigerian hospitals, without transparent performance metrics or local validation. Data from Nigerian patients could be extracted to remote servers for model training, with limited safeguards or benefits returning to local health systems. Public-private partnerships might prioritize high-margin dialysis services over population-level prevention and early detection efforts.

To mitigate these risks and harness AI for public benefit, the analysis suggests several governance principles for Nigerian kidney health initiatives:

Local validation and transparency. Any AI tool used in clinical decision-making should be validated on Nigerian patient data, with performance metrics stratified by key subgroups (e.g., sex, age, region). Model logic and limitations should be documented and accessible to clinicians and regulators.

Data sovereignty and protection. Kidney registries and CKD surveillance databases should operate under clear data protection laws. Data sharing agreements must specify permissible uses, anonymization standards, and prohibitions on secondary exploitation without consent.

Equity-oriented design. Algorithms should be evaluated for differential performance across socio-economic and geographic groups. Regulatory bodies should require evidence that AI tools do

not exacerbate existing inequities in access or outcomes.

Participatory governance. Nephrology societies, primary care representatives, patient advocacy groups, and data scientists should participate in the development of guidelines and standards for AI and data use in kidney care.

These principles align with Birhane et al.'s (2026) call for resisting regulatory capture and embedding structural justice in AI governance. In the kidney health domain, they would guide the design of a "Kidney Health Data Charter" and specific regulatory instruments under Nigeria's health and data protection laws.

5. An integrated Kidney Health Intelligence Framework for Nigeria

Bringing together the above strands, the analysis yields a proposed Kidney Health Intelligence Framework with four interacting pillars:

1. Population risk surveillance. Adapted outbreak detection methods to monitor CKD-related indicators, informing targeted screening and prevention.
2. AI-enabled clinical decision support. Transfer learning-based risk prediction tools and decision aids embedded in primary and secondary care workflows.
3. Interdisciplinary care networks. Multidisciplinary teams spanning nephrology, primary care, nursing, nutrition, social work, and data analytics, coordinated through shared registries and care pathways (Interdisciplinary, n.d.; This, n.d.).
4. Governance and regulation. A Kidney Health Data Charter, AI validation standards, and participatory oversight mechanisms aligned with national health policies and data protection laws (Birhane et al., 2026).

Information flows between these pillars are bidirectional: surveillance data feed into clinical decision support model updates; clinical registries inform surveillance dashboards; interdisciplinary teams interpret and act on data signals; and regulatory bodies oversee data use and AI tool deployment. In essence, the framework reframes kidney failure not as an inevitable outcome of individual risk factors, but as a system-level intelligence and coordination problem amenable to interdisciplinary interventions.

Discussion

The analysis suggests that interdisciplinary approaches from adjacent fields can plausibly improve outcomes related to the rising incidence of kidney failure in Nigeria. By integrating informatics, public health surveillance, interdisciplinary care, and regulatory governance, Nigeria could shift from a predominantly reactive, tertiary-care-oriented kidney health model toward a proactive, data-informed, and equity-oriented system.

Implications for clinical practice and service organization

For clinicians and health managers, the Kidney Health Intelligence Framework underscores the importance of early detection and continuity of care. Simple AI-assisted risk stratification tools deployed in primary care, integrated with national hypertension and diabetes programs, could increase CKD detection at earlier stages, enabling more aggressive blood pressure control, glycemic management, and lifestyle interventions (Hypertension, n.d.; In, n.d.). Interdisciplinary care teams, as advocated in the CKD literature, can address the multifaceted needs of patients, from medication management to psychosocial support (Interdisciplinary, n.d.; This, n.d.).

Importantly, these interventions need not rely on cutting-edge technology alone. Many benefits can be realized through low-tech adaptations: paper-based risk score charts; simple registries tracking CKD patients at the facility level; and regular multidisciplinary case conferences. AI and deep learning become most valuable when layered onto solid clinical and data foundations, enhancing rather than substituting for good clinical practice.

Public health and health system implications

From a public health perspective, treating CKD like a slowly evolving "outbreak" highlights the need for surveillance, prevention, and rapid response capabilities. Integrating CKD indicators into existing health information systems, such as the District Health Information System commonly used in African countries, could allow managers to monitor trends and evaluate the impact of interventions (Babanejaddehaki et al., 2024). Public health campaigns targeting hypertension awareness, salt reduction, and safe use of traditional medicines can be better targeted using surveillance data that identify high-risk communities (Grandville Medical And Laser Dr, n.d.; Authors unavailable, n.d.-a).

Moreover, framing kidney failure as a system problem supports advocacy for strengthened primary care, health financing reforms (e.g., inclusion of CKD services in national health insurance benefits), and cross-sectoral policies addressing structural determinants such as urban planning and food systems (In, n.d.; Several, n.d.). Surveillance findings can be powerful tools for mobilizing political will and resources when communicated effectively.

Regulatory and ethical considerations

However, the introduction of AI and intensified data collection heightens the importance of robust regulatory and ethical frameworks. Nigerian policymakers must anticipate and prevent forms of regulatory capture in which technology vendors disproportionately shape AI adoption pathways (Birhane et al., 2026). Lessons from other domains-such as the rapid, sometimes uncritical adoption of algorithmic credit scoring or surveillance technologies-warn against techno-solutionism detached from local needs and oversight capacity.

Ethical concerns include privacy, informed consent for secondary data use, and the risk of

exacerbating regional inequities if AI tools are implemented first in wealthier urban hospitals. To address these issues, Nigeria could establish a multi-stakeholder committee on AI and data in health, with a specific sub-group focusing on non-communicable diseases, including CKD. This body could develop guidelines on local validation, data governance, and equity assessments for any AI tools deployed in nephrology.

Cooperation and implementation challenges

Drawing on Vallstrom's (2024) analysis of cooperation under evolutionary pressure, the success of the Kidney Health Intelligence Framework depends on aligning incentives and fostering trust among actors. Potential barriers include professional turf wars (e.g., between nephrologists and primary care providers), data ownership disputes, and resource constraints. Addressing these requires deliberate design of cooperative structures: shared performance metrics across levels of care; transparent data sharing agreements; and participatory forums where clinicians, managers, and patients co-create protocols.

The suggested 18-month implementation and research timeline embedded in the original outline provides a pragmatic roadmap:

Months 1-3: Detailed literature review; stakeholder mapping; co-design of pilot surveillance and decision support components with frontline clinicians and data teams.

Months 4-7: Pilot data collection in selected facilities and states; development and calibration of basic risk prediction models; establishment of a minimal CKD registry.

Months 8-10: Iterative refinement of models and dashboards; training of interdisciplinary teams; initial evaluation of feasibility and acceptability.

Months 11-12: Synthesis of findings into a refined Kidney Health Intelligence Framework; policy briefs and recommendations.

Months 13-15: Dissemination via workshops with federal and state health authorities, professional associations, and patient groups; planning for scale-up where appropriate.

Months 16-18: Follow-up monitoring of early implementation; assessment of process outcomes (e.g., increase in early CKD detection, improved blood pressure control among CKD patients); identification of barriers and enablers for further research.

This phased approach embodies cooperative, adaptive learning. It allows for incremental improvements and course corrections, reducing the risk of costly missteps.

Limitations and future research

The conceptual nature of this study means that its conclusions require empirical validation in Nigerian settings. Future research should include:

Prospective cohort studies to refine CKD risk prediction models using Nigerian data.

Implementation research on integrating CKD screening and management into primary care, with and without AI-based decision support.

Pilot CKD surveillance projects employing adapted outbreak detection methods, with evaluation of their impact on early detection and resource allocation.

Policy analysis and qualitative research on regulatory capacity for AI in health, focusing on kidney care as a test case.

Economic evaluations of interdisciplinary kidney care models, including cost-effectiveness of AI-enabled interventions compared to standard care.

Such studies would strengthen the evidence base for or against the proposed interdisciplinary strategies and guide scalable policy adoption.

Conclusion

Kidney failure in Nigeria exemplifies a complex, systemic health challenge at the intersection of biomedical, behavioral, and structural determinants. The rising incidence and high mortality are not inevitable; they reflect gaps in surveillance, early detection, chronic disease management, and governance. This paper has argued that interdisciplinary approaches drawing on health informatics, deep learning, public health surveillance methods, and regulatory studies can significantly improve outcomes along the kidney failure continuum.

By adapting outbreak detection techniques for CKD surveillance, leveraging transfer learning for risk prediction and diagnostics, organizing interdisciplinary care networks, and instituting robust, equity-oriented regulatory frameworks, Nigeria can build a Kidney Health Intelligence Framework capable of detecting risk earlier, coordinating responses more effectively, and protecting patients' rights and interests. Realizing this potential will require cooperative action among clinicians, public health officials, technologists, regulators, and communities, as well as sustained investment in data systems and capacity building.

Ultimately, the question is not only whether interdisciplinary approaches can improve kidney failure outcomes in Nigeria, but whether stakeholders will seize the opportunity to reconfigure kidney health as a domain of intelligent, just, and collaborative care. The conceptual framework and roadmap presented here are intended as a starting point for that collective endeavor.

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